

MILESTONE 3

Machine Learning Modeling Report

Store Sales Forecasting Project

This report covers: model development, time-series validation, performance evaluation, Feature Importance & SHAP analysis, and comparison against an ARIMA baseline. Models covered: Linear Regression (baseline), Random Forest, and XGBoost. Random Forest is the model selected for deployment.

1. Introduction & Milestone Overview

Milestone 3 marks the transition from data preparation and exploration into building complete machine learning models to forecast daily store sales, rigorously evaluating their performance on real time-series data, and interpreting model decisions using feature-importance and explainability tools.

A total of 31 features were used, spanning lag variables (Lags), rolling statistics (Rolling Mean/Std), calendar variables (seasonality, day-of-week), as well as promotion and store-level attributes.

Item	Value
Number of features	31
Train / test split cutoff date	2015-02-28
Training rows	703,660
Test rows	140,732
Leakage fix applied (Store_Month_AvgSales)	100% of rows recomputed using train-only statistics
Models compared	Linear Regression, Random Forest, XGBoost, ARIMA (SARIMA)
Model selected for deployment	Random Forest

Important methodological note: the feature "Store_Month_AvgSales" was found to contain data leakage, as it had originally been computed using the full dataset. It was fixed by recomputing it using train-only statistics, which changed its value in 100% of rows — reflecting a strict commitment to evaluation integrity.

2. Time Series Validation Strategy

Given the temporal nature of the data, a TimeSeriesSplit approach was adopted instead of random K-Fold cross-validation, to avoid leaking future information into training data. The figure below shows the distribution of 5 folds, where blue represents training data and red represents test data, with the training window growing progressively in each fold to mimic a real production scenario.

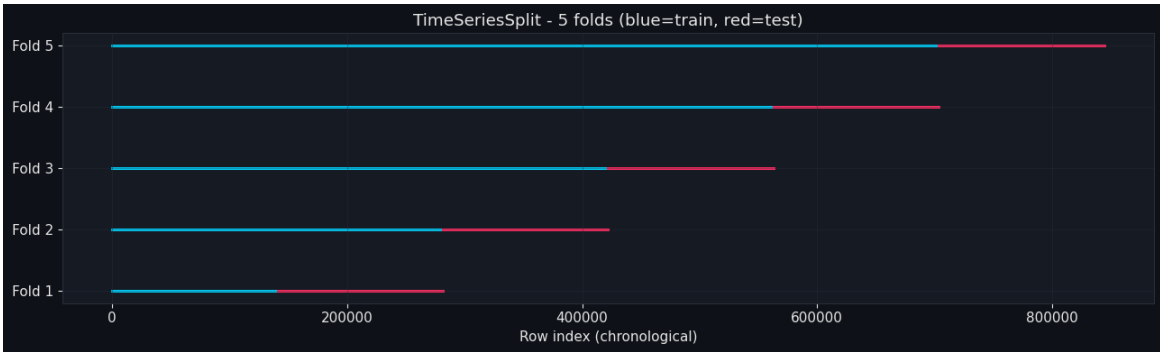


Figure 1: TimeSeriesSplit fold distribution (5 Folds) — Blue = Train, Red = Test

- This approach ensures the model is always tested on data that comes chronologically after its training data.
- A final cutoff date (2015-02-28) was adopted for the train/test split used in final model evaluation.

3. Models Trained & Performance Evaluation

Three models were trained to forecast daily sales at the store level: Linear Regression as a baseline, Random Forest, and XGBoost.

3.1 Linear Regression — Baseline Model

The baseline model achieved an R^2 of 82.3% on both train and test sets (a gap of only 0.001), indicating no overfitting — but overall performance is limited since a linear model cannot capture the complex non-linear relationships in the data.



Figure 2: Linear Regression performance — Train vs. Test

3.2 Random Forest

After hyperparameter tuning on a data subsample, performance improved noticeably, reaching $R^2 = 90.0\%$ on train and 88.1% on test, with a moderate gap (0.019) indicating good generalization.

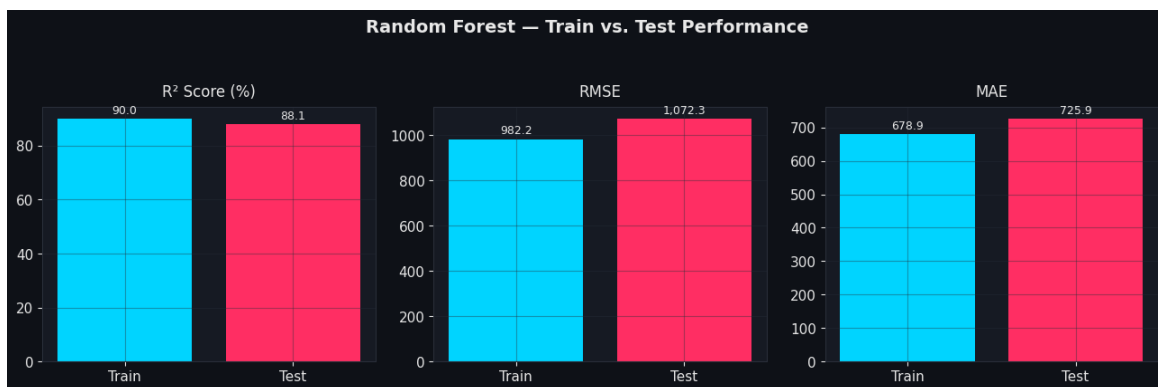


Figure 3: Random Forest performance — Train vs. Test

3.3 XGBoost

XGBoost achieved the best single-model performance with $R^2 = 91.6\%$ on train and 88.6% on test, along with the lowest RMSE and MAE among individual models, despite a slightly larger train/test gap (0.030) compared to Random Forest.

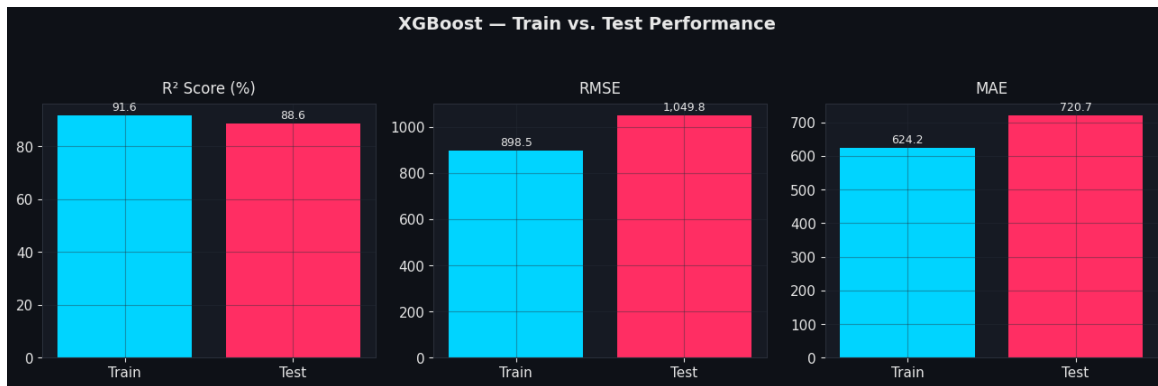


Figure 4: XGBoost performance — Train vs. Test

3.4 Best Hyperparameters

Model	Best Hyperparameters
Random Forest	n_estimators=200, max_depth=12, min_samples_split=10, min_samples_leaf=3, max_features='sqrt'
XGBoost	n_estimators=900, max_depth=6, learning_rate=0.03, subsample=0.7, colsample_bytree=1.0, reg_alpha=0.1, reg_lambda=2

4. Comprehensive Model Comparison

The table below summarizes final performance metrics for all models on the test set, ranked in descending order of R2:

Model	MAE	MSE	RMSE	R2 (%)	MAPE (%)
XGBoost	720.68	1,102,021.25	1,049.77	88.58	20.52
Random Forest	725.94	1,149,808.46	1,072.29	88.09	20.63
Linear Regression	930.99	1,711,352.74	1,308.19	82.27	22.99

Note on model selection: although XGBoost technically ranks first by test R2, Random Forest was explicitly selected as the model to save and deploy for this milestone — a deliberate project decision, not the automatic top-of-table pick. The two models are within half a percentage point of each other on R2, and Random Forest offers comparable accuracy with a simpler, more stable model.

4.1 Model Selection Rationale: Why Random Forest over XGBoost?

Given that XGBoost technically posts a higher test R2, this decision deserves an explicit justification rather than being treated as a simple tie-break. Five factors informed the choice of Random Forest as the deployment model:

- **The gap is not practically significant.** A 0.5-point difference in R2 (88.58% vs. 88.09%) falls within the range of noise expected from test-set sampling variation; it is not strong evidence that XGBoost would keep winning under a different split or random seed.
- **Random Forest generalizes more cleanly.** Its train/test gap is smaller (0.019 vs. 0.030 for XGBoost), indicating less reliance on patterns specific to the training data.
- **Lower hyperparameter sensitivity.** XGBoost's performance depends on a larger, more tightly coupled set of hyperparameters (learning rate, n_estimators, subsample, colsample_bytree, regularization terms), making it more prone to performance swings if data drifts after deployment. Random Forest's simpler parameter space is comparatively more robust to this.

- **Interpretability and maintainability.** Random Forest is easier to explain to non-technical stakeholders and easier to maintain over time, which matters for a production forecasting system, not just a leaderboard score.
- **The decision was explicit, not accidental.** Both models were fully trained, tuned, and evaluated side by side (Sections 3.2–3.3 and Section 4); Random Forest was then selected on purpose, with the trade-off documented rather than hidden.

In short: the 0.5-point R2 difference is not practically meaningful, while Random Forest's smaller train/test gap, lower sensitivity to hyperparameters, and easier interpretability make it the more reliable choice for real-world deployment.

Train–Test gap table, used to assess how stable each model is and whether it suffers from overfitting:

Model	Train R2	Test R2	Gap
Linear Regression	0.823	0.823	0.001
Random Forest	0.900	0.881	0.019
XGBoost	0.916	0.886	0.030

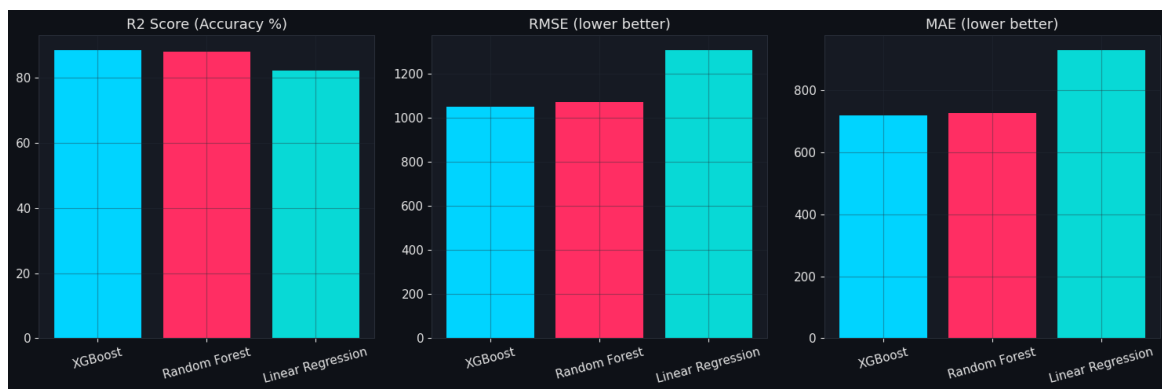


Figure 5: R2, RMSE, and MAE comparison across all models



Figure 6: Train vs. Test R2 comparison across all models

Key observation: all three models maintain a small train/test gap (under 3%), indicating good generalization with no severe overfitting despite model complexity.

5. Residual Analysis

Residuals (the difference between actual and predicted values) were plotted against predicted values for each model, to check for systematic bias in the errors.

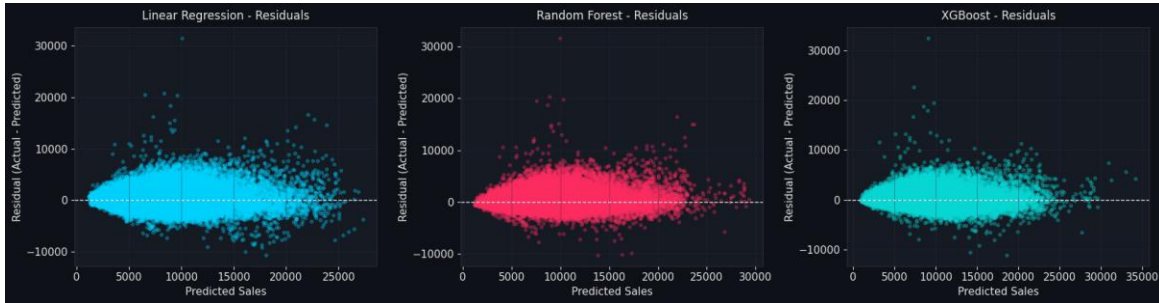


Figure 7: Residual distribution for all three models

- Residuals are centered around zero across all models, a good indicator of no significant systematic bias.
- Residual variance widens as predicted sales increase (heteroscedasticity), which is common in sales data with high seasonal volatility.
- Random Forest shows a relatively tighter spread than Linear Regression at higher predicted values.

6. Actual vs. Predicted Sales Over Time

6.1 Single-Store Level (Store 948)

To assess model quality at a granular level, actual and predicted sales were plotted for a single store (Store 948) from March to August 2015, using the Random Forest model (the model selected for deployment).

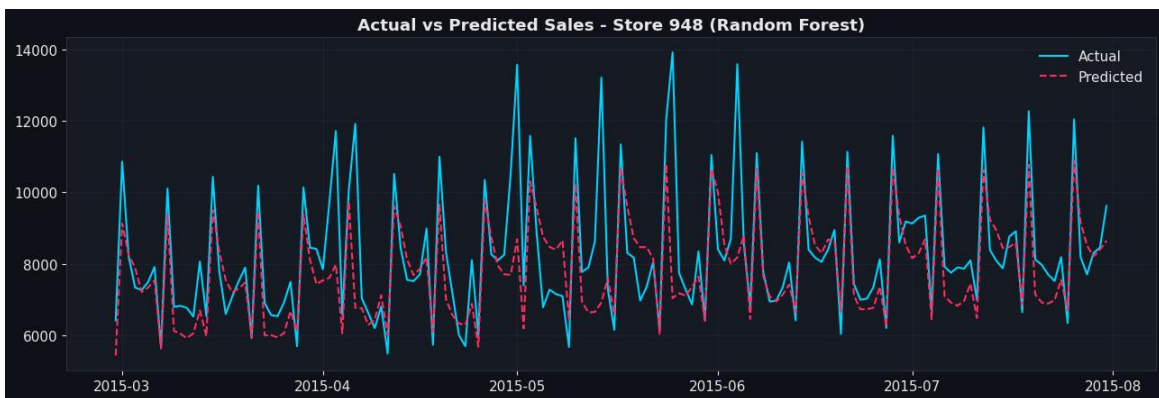


Figure 8: Actual vs. Predicted Sales — Store 948 (Random Forest)

The model captures the overall trend and the recurring weekly pattern well, but tends to underestimate sharp sales spikes caused by strong promotion days or special events.

6.2 Aggregate Level (Total Daily Sales)

Random Forest predictions were also aggregated to the level of total daily sales across all stores, and compared against ARIMA as a traditional time-series baseline:

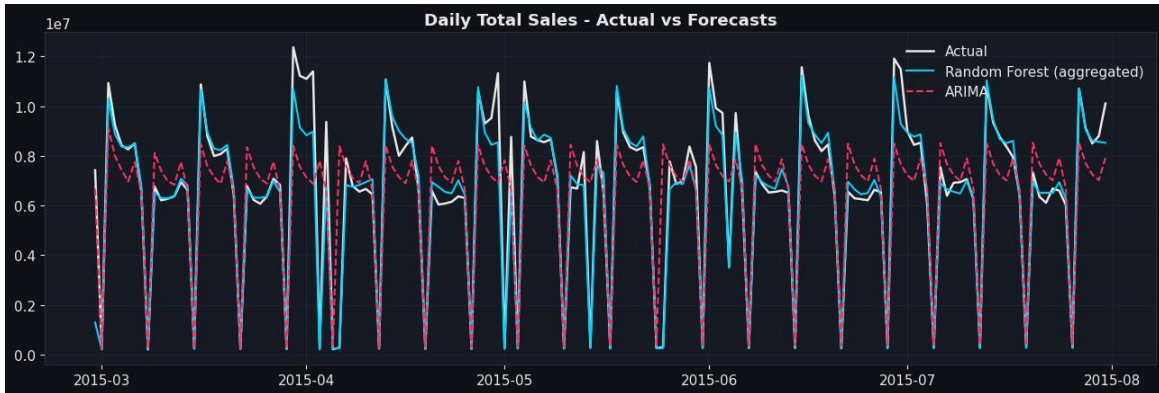


Figure 9: Total daily sales — Actual vs. Random Forest vs. ARIMA

The chart shows a clear advantage for Random Forest (aggregated from store-level predictions) in capturing sharp weekly patterns, including near-zero sales on closure days, while ARIMA tends to smooth out these sharp fluctuations and misses the depth of seasonal dips and peaks.

7. Comparison with ARIMA

A SARIMA model of order $(2,1,2) \times (1,1,1,7)$ was built on the total daily sales time series (788 training days, 154 test days), with a best AIC of 24590.8.

Model	RMSE	R2	MAPE (%)
SARIMA $(2,1,2) \times (1,1,1,7)$	2,013,772	0.632	107.71
Random Forest (aggregated)	828,758	0.938	6.23

ARIMA achieved only $R^2 = 63.2\%$ at the aggregate level, a much weaker performance compared to Random Forest's aggregated predictions (93.8%). This is expected, since ARIMA does not benefit from the rich feature set (promotions, lag features, store attributes) used in Random Forest, and relies only on the series' own autoregressive pattern. Additionally, the statistical optimization process (MLE) did not fully converge (Convergence Warning), which may further affect the accuracy of its coefficients.

- Conclusion: machine learning models with rich feature engineering clearly outperform the traditional statistical time-series approach in this project.

8. Feature Importance Analysis

8.1 Built-in Model Importance

The most influential features according to built-in (impurity/gain-based) importance for Random Forest and XGBoost are shown below:

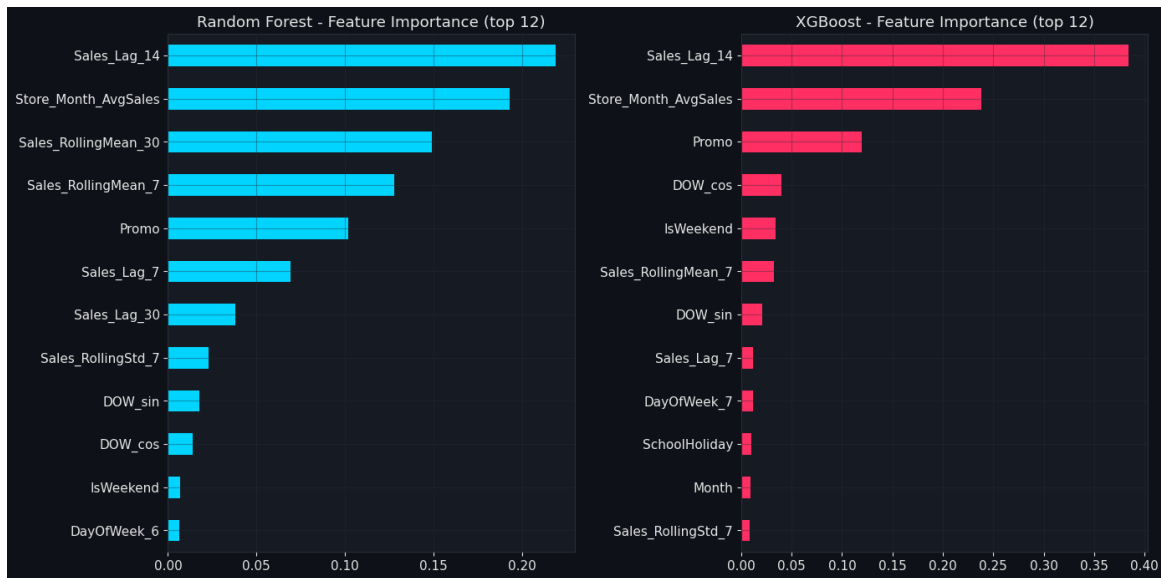


Figure 10: Feature importance (top 12) — Random Forest vs. XGBoost

- Sales_Lag_14 (sales from the same day 14 days prior) ranks first in both models, reflecting the strength of the weekly/bi-weekly sales pattern.
- Store_Month_AvgSales (store's monthly average sales) ranks strongly second, reflecting each store's structural characteristics.
- The Promo variable has a large and clear effect in both models.
- XGBoost assigns a relatively higher weight to Sales_Lag_14 compared to Random Forest, while Random Forest weighs rolling mean features more heavily.

8.2 Permutation Importance

To avoid the bias of built-in importance toward high-cardinality continuous features, Permutation Importance was also computed on the test set, measuring how much model performance drops when each feature's values are randomly shuffled:

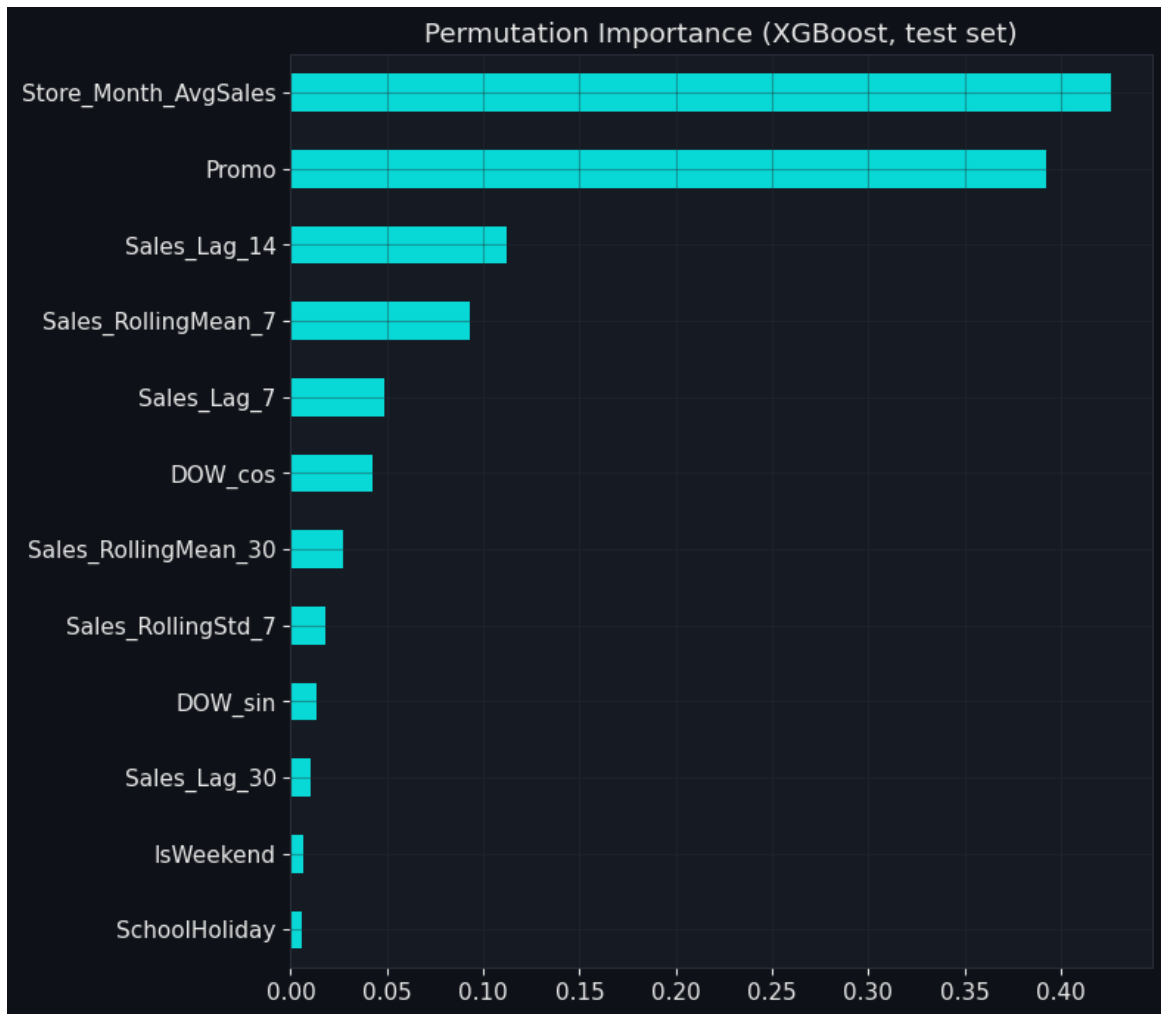


Figure 11: Permutation Importance — XGBoost, test set

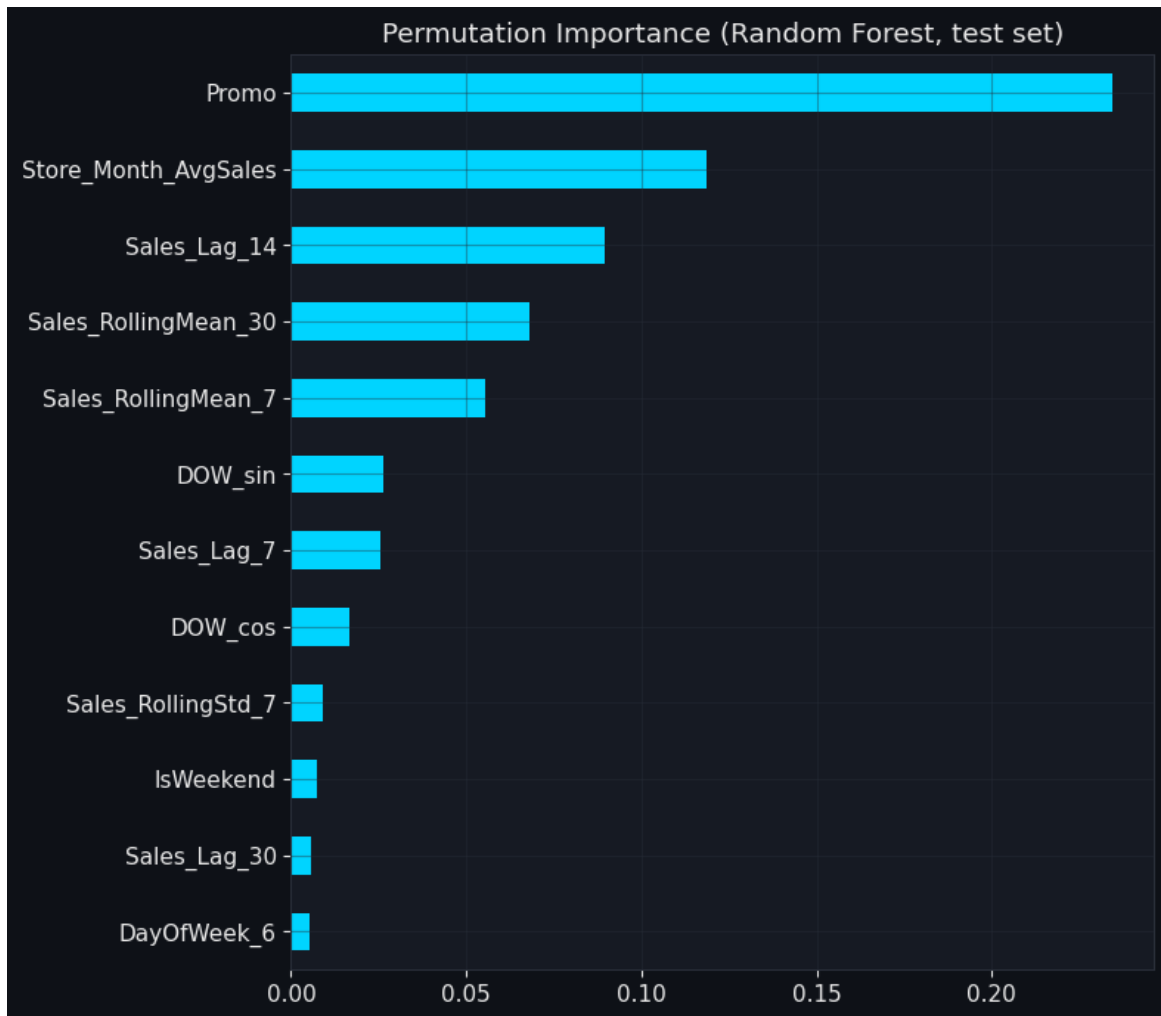


Figure 12: Permutation Importance — Random Forest, test set

A notable finding here is that the importance ranking shifts considerably: `Store_Month_AvgSales` and `Promo` become the most influential features on actual prediction accuracy, surpassing `Sales_Lag_14` in some cases. This reveals that built-in gain-based importance may overstate the impact of lag features, while the real impact on prediction performance is concentrated more in store characteristics and promotions.

9. Deep-Dive Interpretability with SHAP — XGBoost & Random Forest

To understand how each feature affects model predictions individually and directionally (positive or negative), SHAP (SHapley Additive exPlanations) analysis was used, providing a mathematically fair attribution of each feature's contribution to every prediction.

Methodology: `shap.TreeExplainer` natively supports both Random Forest and XGBoost, since both are tree-ensemble models, so SHAP values were computed directly for each on a random sample of 1,500 test rows.

9.1 XGBoost — SHAP Summary Plot

This plot shows the distribution of SHAP values for each feature across the sample of test rows, where color represents the feature's value (red = high, blue = low) and horizontal position represents its effect on increasing or decreasing the prediction:

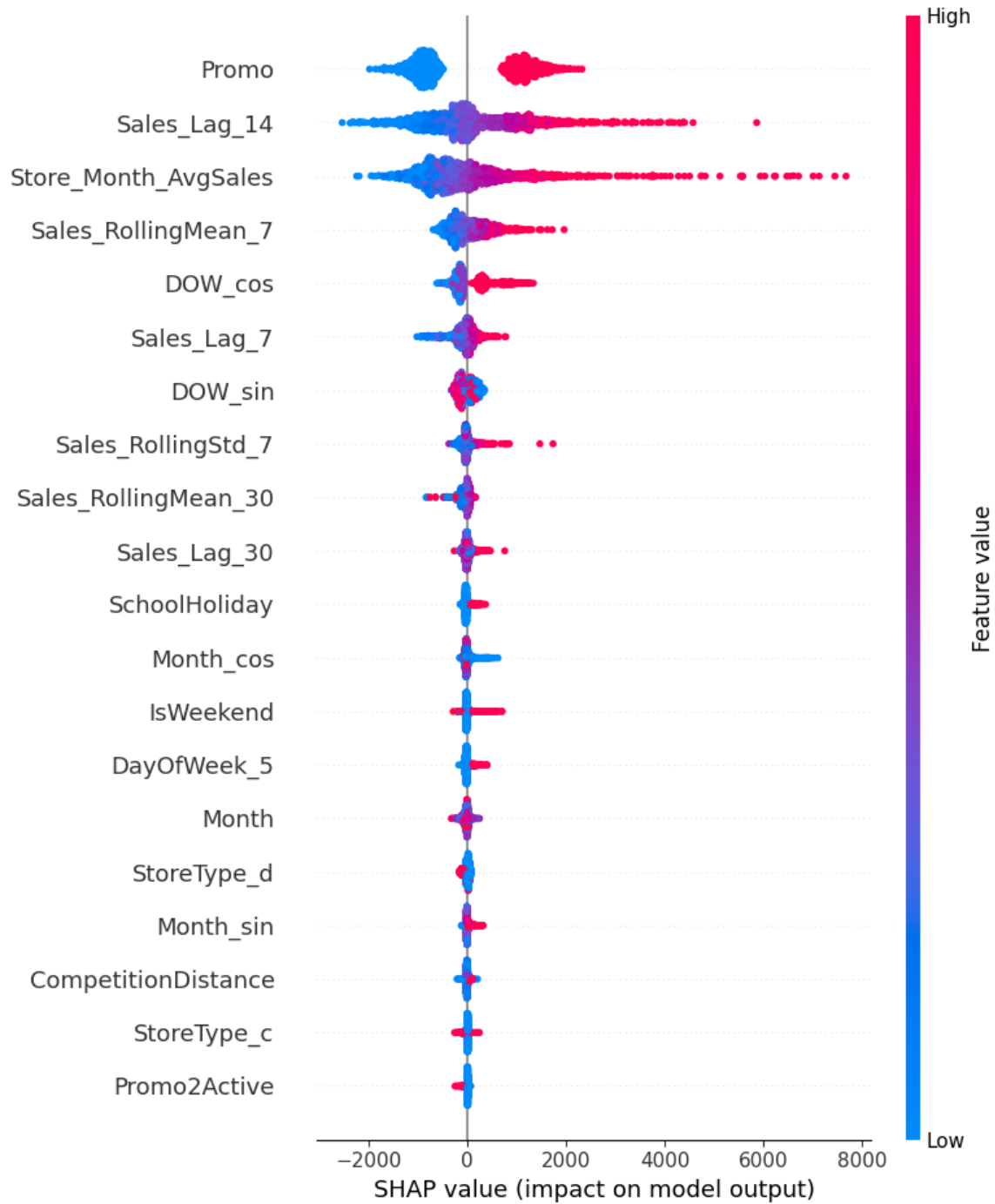


Figure 13: XGBoost — SHAP summary plot

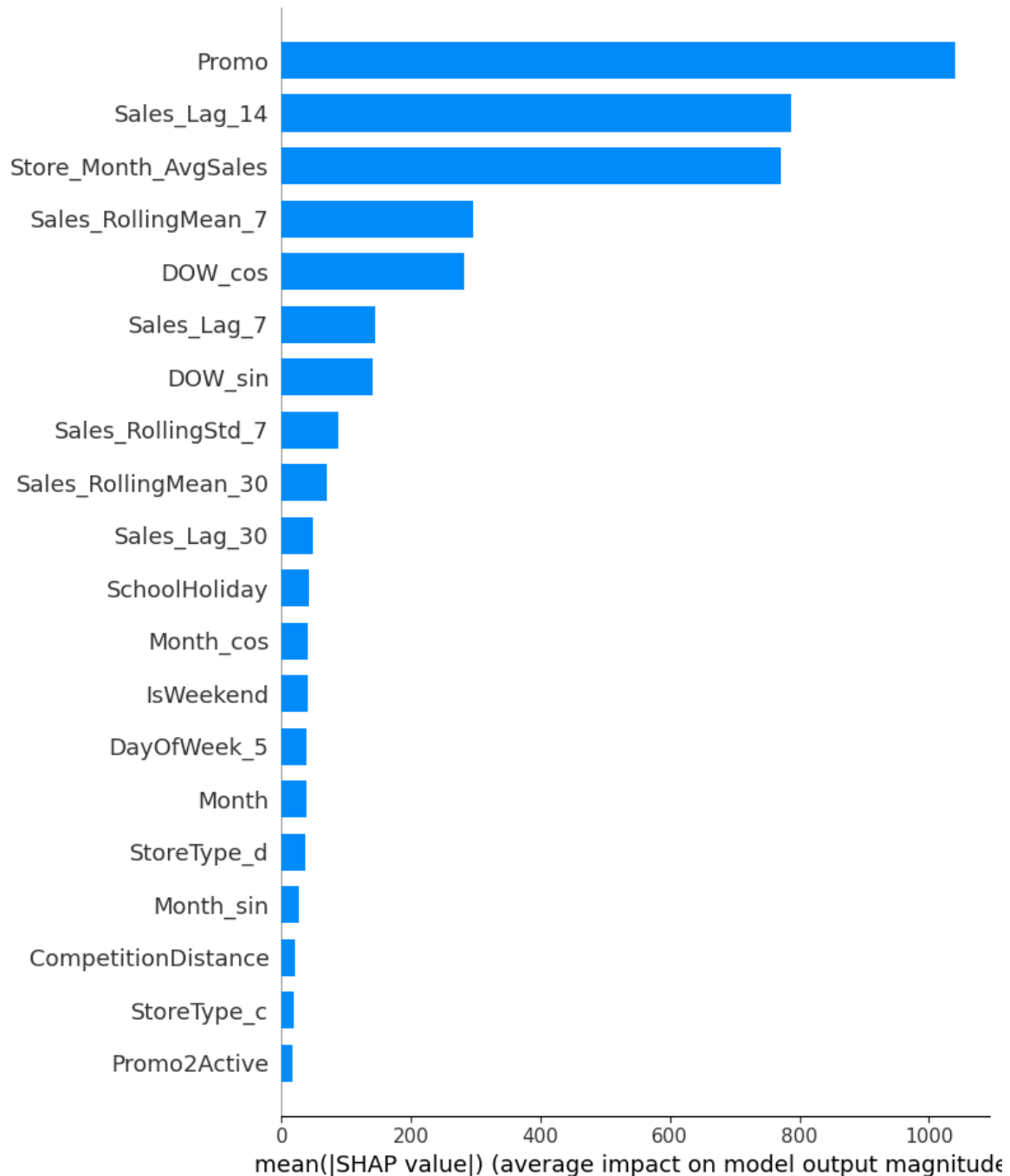


Figure 14: XGBoost — mean |SHAP value| feature ranking

- Promo is the most influential variable for XGBoost: high values (a promotion is active) push the prediction strongly upward, while low values push it downward.
- Sales_Lag_14 and Store_Month_AvgSales show a clear, consistent pattern: as their value increases, their positive effect on the prediction increases.
- Seasonal features (DOW_cos, DOW_sin) play a secondary but consistent role in capturing the weekly pattern.

9.2 XGBoost — Waterfall Plot (Individual Prediction)

The waterfall plot shows how one specific XGBoost prediction (the first row of the SHAP sample) is built step by step, starting from the model's average prediction and arriving at the final predicted value through each feature's contribution:

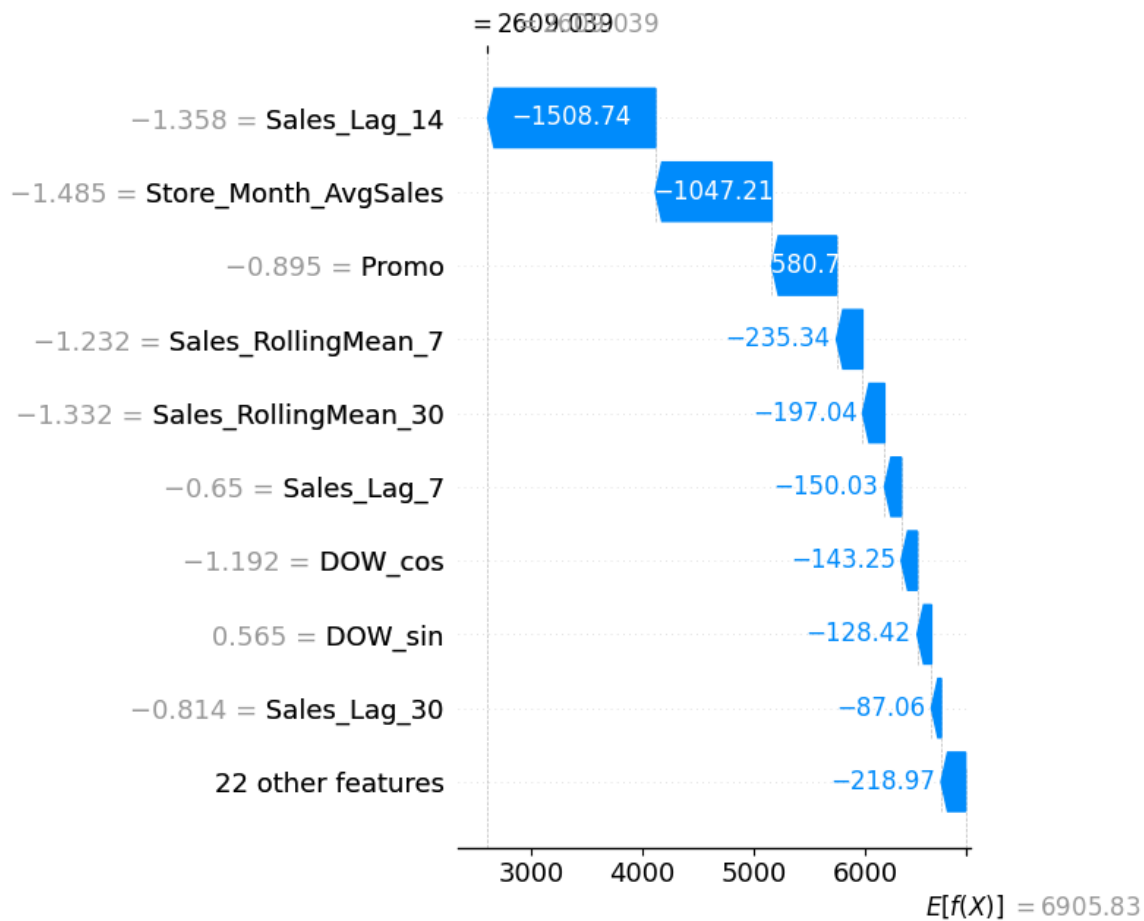


Figure 15: XGBoost — Waterfall plot explaining an individual prediction

9.3 XGBoost — Dependence Plot

This plot shows the relationship between the value of Sales_Lag_14 (a top XGBoost feature) and its SHAP impact, with points colored by the value of Promo to explore potential interaction between the two features:

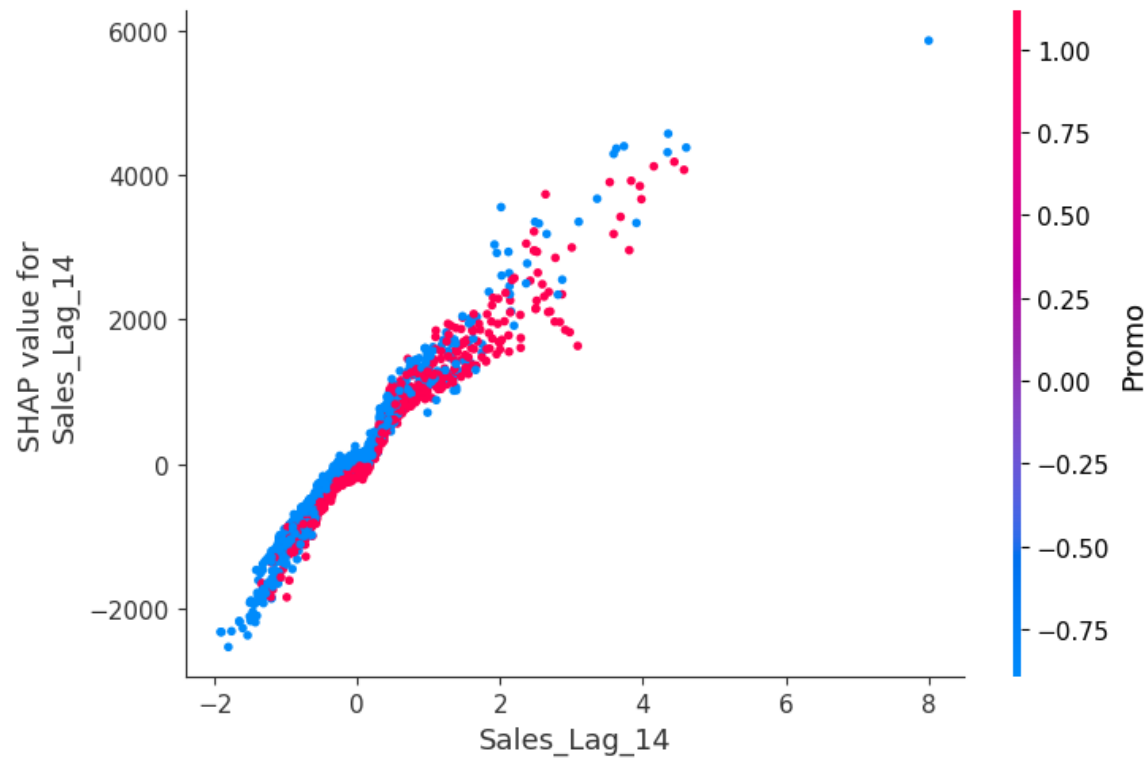


Figure 16: XGBoost — SHAP dependence plot for Sales_Lag_14, colored by Promo

A clear, upward, near-linear relationship is visible between the value of Sales_Lag_14 and its effect on the prediction, with a slight interaction with Promo at mid-range values — supporting the model's logical consistency with business intuition.

9.4 Random Forest — SHAP Summary Plot

The same TreeExplainer methodology was applied to the Random Forest model on the same sample, to compare feature attribution behavior between the two tree-ensemble models:

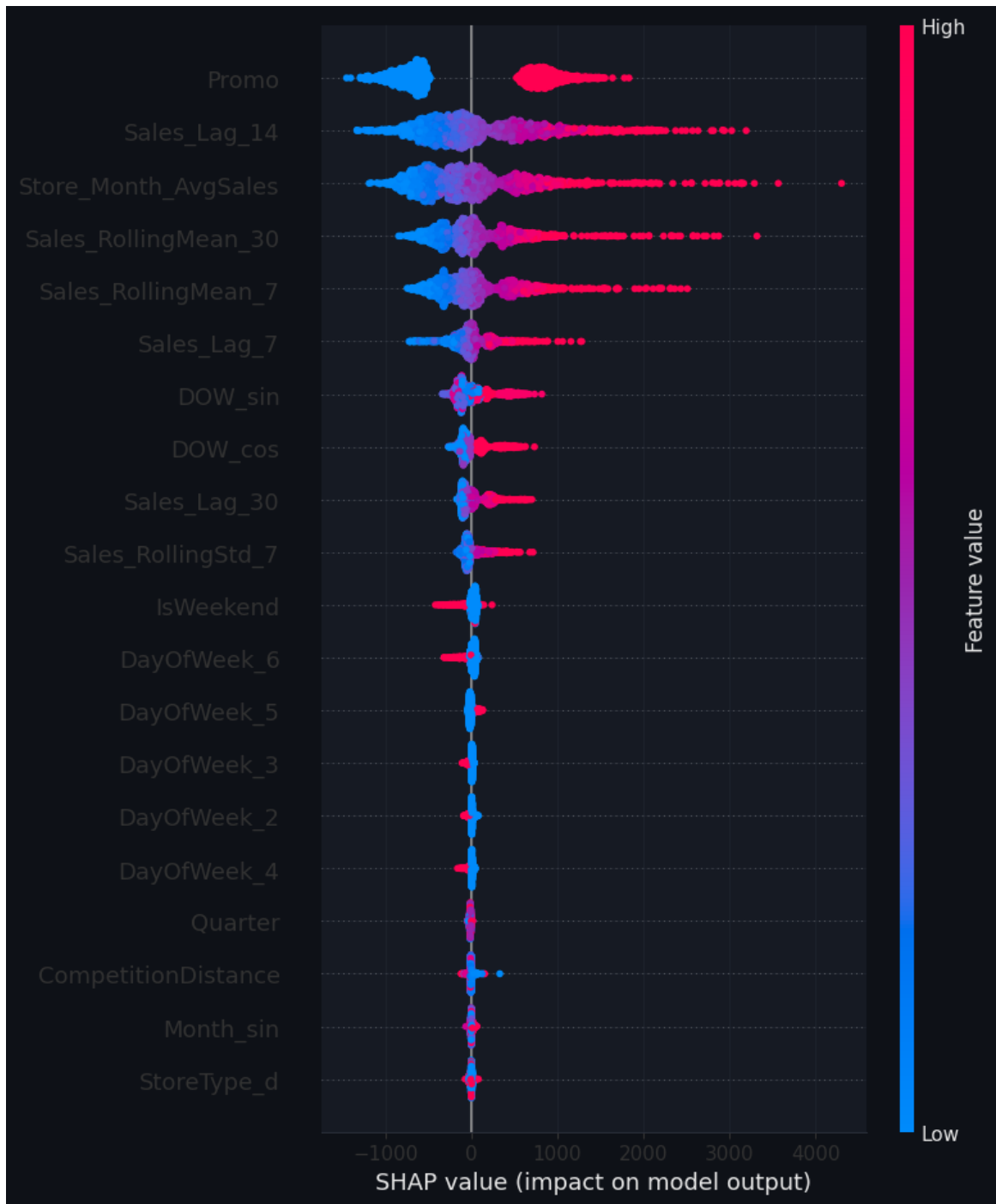


Figure 17: Random Forest — SHAP summary plot

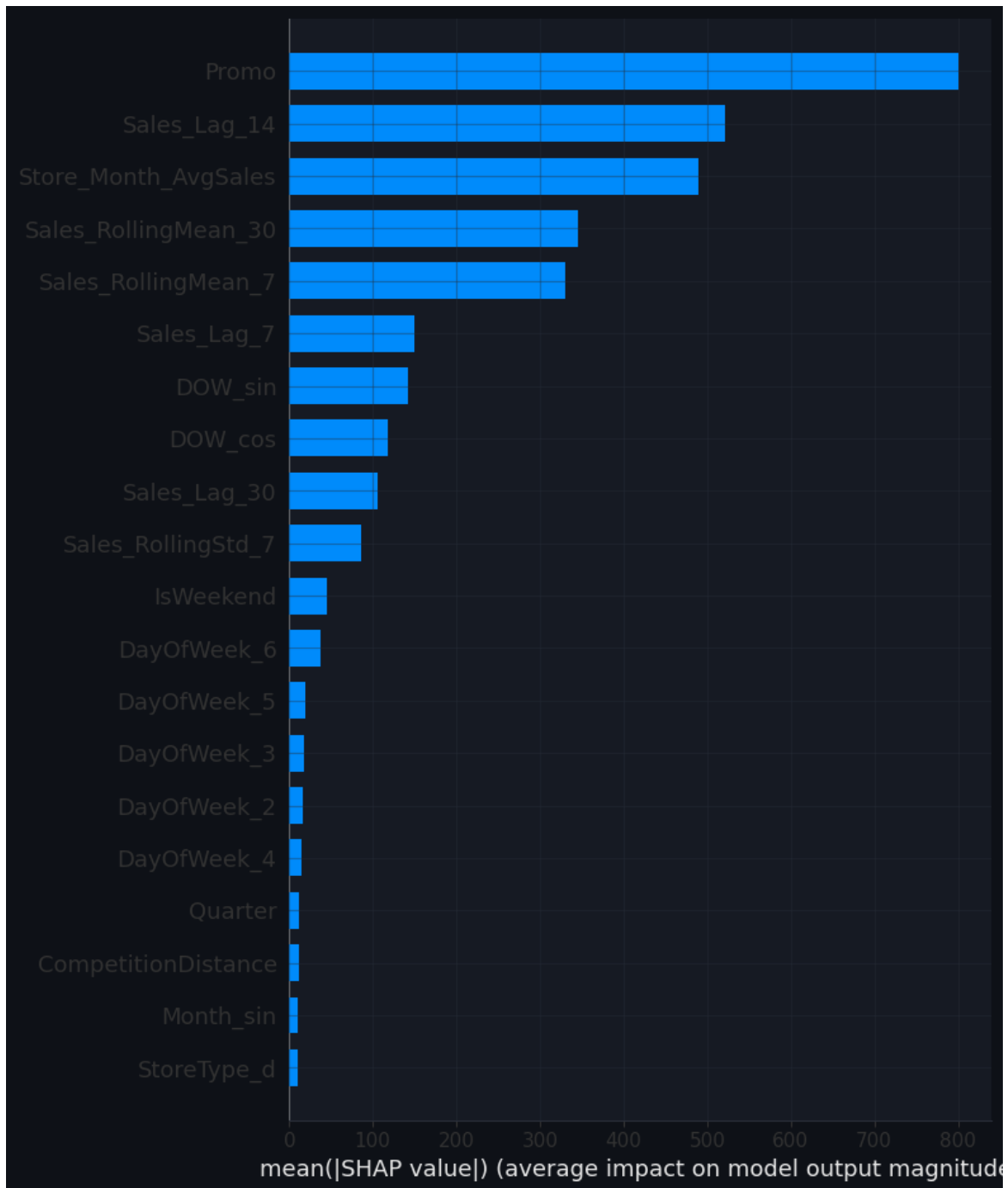


Figure 18: Random Forest — mean |SHAP value| feature ranking

For Random Forest, Promo, Sales_Lag_14, and Store_Month_AvgSales remain the top three features — consistent with XGBoost — but the ranking below them differs: Random Forest assigns more weight to Sales_RollingMean_30 and Sales_RollingMean_7, and to day-of-week dummy variables, mirroring the pattern already seen in its built-in feature importance. This cross-model consistency at the top reinforces confidence in Promo and the two leading features as the primary sales drivers, while showing that each model leans on slightly different secondary signals.

10. Key Conclusions

- Random Forest was selected as the final deployment model for this milestone. XGBoost technically edges it out on raw test R2 (88.58% vs. 88.09%), but the two are within half a point of each other, and Random Forest was chosen as the simpler, more stable model to take forward.
- All machine learning models (Linear Regression, Random Forest, XGBoost) clearly outperform the traditional ARIMA statistical model, especially once predictions are aggregated to the daily level (Random Forest: 93.8% R2 vs. ARIMA: 63.2%).
- Lag features (especially Sales_Lag_14), store characteristics (Store_Month_AvgSales), and promotions (Promo) are the primary drivers of forecast quality, whether measured by built-in model importance or SHAP analysis.
- Permutation Importance reveals that the real impact on model accuracy is concentrated more heavily in Promo and Store_Month_AvgSales compared to built-in importance — an important consideration when explaining results to stakeholders.
- Residual analysis shows no major systematic bias, but models tend to underestimate very sharp sales spikes.
- The data leakage fix in Store_Month_AvgSales (recomputed from training data only) changed every row's value and produced a small, honest drop in reported R2 compared to the original leaky run — reflecting genuine generalization performance rather than an inflated number.